Mathematical Stories

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Lecture Notes

July 14, 2026

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Introduction

Purpose and Philosophy

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognises the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, formulate precise questions, distinguish an object from its notation, notice structure, and build simple but meaningful chains of reasoning. A solved problem is valuable not only because it produces an answer, but also because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course is organised into three connected parts: linear algebra, analytic geometry, and calculus. In the first part, we focus on matrix methods: matrices, determinants, inverse matrices, and systems of linear equations. The second part develops the language of coordinates and vectors and uses it to describe points, lines, planes, directions, distances, and angles. The third part introduces sequences and functions, limits, continuity, derivatives, integrals, change, and accumulation. These parts are not isolated collections of techniques. Together they form a basic mathematical language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorise, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as an answer to a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some habits of thought will deliberately return in different places. The same question of what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. A mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognise patterns of reasoning rather than only patterns of exercises.

The number of answers produced is not, by itself, a measure of progress. What matters is the quality of the work. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each important step was valid? Did we interpret or verify the result? These questions are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as the art of asking good questions and giving well-reasoned answers.

How the Course Works

This book is part of a weekly learning process rather than a collection of material to be read only before an examination. Each of the twelve thematic chapters is designed to support one week of study. A chapter provides the mathematical structure for the lecture, a complete narrative to which the student can return, and a starting point for individual problem solving and classroom discussion.

In a fifteen-week semester, the twelve thematic weeks establish the main rhythm of the course. The remaining time can be used to introduce the method of work, connect topics, revise important ideas, and present more substantial solutions.

The Weekly Learning Cycle

A student should not leave a topic behind when the lecture ends. A typical week consists of the following connected stages:

1. **Attend the lecture.** The lecture introduces the topic, motivates the central questions, and demonstrates how the relevant mathematical objects and methods should be read.

2. **Return to the chapter.** Read the corresponding chapter carefully after the lecture. The PDF is not merely a summary of slides. It is an organised narrative containing definitions, explanations, examples, interpretations, and mathematical arguments that can be examined more than once.

3. **Ask questions and clarify the material.** Mark every place where a definition, transition, calculation, or conclusion is not yet clear. Use the text, your own examples, classroom discussion, and appropriate digital tools to resolve these questions.

4. **Solve the complete assigned problem set.** Move from following an explanation to constructing arguments independently. Work on every assigned problem and prepare a separate, self-contained note for each one.

5. **Prepare professional mathematical notes.** Use precise notation, explanatory sentences, justified transitions, and a clear conclusion. The note should show not only the answer, but also how it follows from the given information, definitions, and earlier results.

6. **Present and discuss your work.** During exercise classes, be prepared to explain a solution, answer questions about its steps, and describe what you learned while preparing it.

These stages form one continuous process. Each stage prepares the next, and the presentation and feedback lead back to further reading, revision, and improved work.

Exercise Classes: Presentation, Feedback, and Revision

Presenting mathematics is part of learning mathematics. It reveals whether an argument has genuinely been understood and whether the student can communicate it clearly. During a presentation, the student should be able to read the notation, explain the role of each important formula, justify the method, and respond to questions without relying on AI to speak on the student's behalf.

Feedback given during exercise classes is intended for the whole group, not only for the student whose work is currently being presented. Everyone should follow the argument and listen carefully to the discussion. By comparing different examples, students learn what a complete solution looks like, which explanations make an argument clear, which steps require justification, and which weaknesses require revision.

Students should not focus only on comments addressed directly to their own work. Listening to feedback on the work of others provides a broader set of examples and develops an operational understanding of the course standards: what makes mathematical work clear, justified, and complete, and what still needs improvement.

The feedback concerns the mathematical work and its results, not the personal worth of the student. What is examined is the quality of the note, the understanding demonstrated in the argument, and the clarity of the presentation. Feedback should identify what already works and what should be changed in order to produce a stronger solution.

The first submitted version does not always have to be the final version. Students should learn to receive comments, reconsider their choices, correct errors, add missing explanations, and improve both the written note and its presentation. Revision is not evidence of failure. It is a normal part of serious intellectual work.

This process is important throughout university study and later in professional life. Students and professionals are given tasks, asked to solve problems, expected to present results, and required to respond to feedback and critical review. Learning to improve an initial solution is therefore not an additional classroom exercise. It is a fundamental habit of responsible academic and professional work.

Progress and Assessment

Progress is demonstrated by the quality and increasing independence of the student's work. By the end of a weekly cycle, the student should be able to identify the main mathematical objects, state the essential definitions, solve representative problems, justify the principal steps, verify or interpret the result, and communicate the argument clearly.

Assessment in the exercise component focuses primarily on how the student works throughout the semester. Evidence of serious engagement includes:

- regular preparation and completion of the assigned problem sets,
- documented attempts to solve problems rather than bare answers,
- active participation in presentations and classroom discussion,
- separate, self-contained notes for individual problems,
- careful revision of work after feedback,
- increasing clarity, independence, and mathematical precision.

A student is not defined by the quality of the first solution or presentation. Some students may initially find mathematical notation, formal writing, or public explanation difficult. A weak beginning does not prevent a very good final grade for the exercise component. A student who works conscientiously, responds to feedback, corrects earlier weaknesses, and demonstrates substantial development may finish the exercise component with a very good grade.

This approach does not lower the standard of mathematical work. It recognises that reaching the standard through sustained effort and revision is itself a meaningful achievement. The exercise-component grade reflects a process of learning, not merely a snapshot taken at the beginning of the semester.

Systematic work according to these guidelines provides the basis for earning a passing grade for the exercise component. The examination is assessed separately and tests the student's mathematical knowledge and ability to use it independently. Regular work does not replace the examination, but it develops the understanding and skills that substantially increase the likelihood of successfully completing the course.

How to Prepare Mathematical Notes

A mathematical note is not merely a place where an answer is recorded. Its purpose is to show how a problem is understood, how relevant information is selected, how a method is chosen, and how a conclusion follows from definitions, assumptions, and earlier results.

Mathematical expressions are compact. A single formula may contain several ideas at once: objects, operations, conditions, and logical relationships. Reading a formula therefore means more than pronouncing its symbols. A student should be able to explain what the objects represent, what operation is being performed, why the expression is meaningful, and what information the formula provides.

The same principle applies to writing. A sequence of formulas without explanatory sentences rarely communicates a complete mathematical argument. Formulas should be connected by words that explain their role. A good solution tells the reader why a calculation is being performed, which definition or theorem is being used, whether its assumptions are satisfied, and how the result answers the original question.

Make the Note Self-Contained

Prepare one separate, self-contained note for each assigned problem, including all of its subparts. Begin by writing the problem or a complete and faithful formulation of it. The reader should not need to search through another document to discover what is being solved. After several weeks, you should still be able to open the note and understand both the question and the solution.

Identify what is given and what must be found, proved, or explained. Introduce the relevant notation and determine what kinds of mathematical objects appear in the problem. Before calculating, ask what information is available and what would count as a satisfactory answer.

Make the Reasoning Visible

Present the principal idea or method before, or while, carrying out the calculation. Important transitions should be explained with complete sentences. For example:

- Because the determinant is non-zero, the matrix is invertible.
- We may therefore multiply both sides by the inverse matrix.
- By the definition of the derivative, we consider the following limit.
- Since the direction vector is non-zero, it determines a line.
- Substituting the result into the original equation verifies the solution.

Such sentences are not decoration around the mathematics. They are part of the mathematics because they show the logical relationship between successive formulas.

It is not necessary to describe every elementary arithmetic operation. The amount of explanation should be proportional to the problem. A simple exercise may require only a few sentences, while a more substantial problem may require a longer argument. The goal is not to make every note long. The goal is to make every essential step understandable and justified.

> Write as briefly as possible, but as completely as necessary.

A Complete Solution

A complete note should make the following five stages visible:

1. **Understand the problem.** Record the problem, identify the given information, and state clearly what must be found, proved, or explained.

2. **Choose a method.** Select the definitions, theorems, or techniques that connect the available information with the goal.

3. **Construct the argument.** Carry out the necessary steps and connect formulas with sentences that explain why each important transition is valid.

4. **Verify the result.** Check calculations, assumptions, domain restrictions, or geometric meaning whenever appropriate.

5. **State the conclusion.** Answer the original question clearly and in its proper context.

A result without its context is not a complete solution. A bare final answer, an unexplained sequence of formulas, or a few unconnected calculations will be treated as incomplete work and should be revised.

Quality cannot be replaced by quantity, but quality does not remove the obligation to complete the assigned work. Students are expected to prepare separate, complete solutions for the entire assigned problem set. Producing many unexplained answers is insufficient, but preparing one polished solution while ignoring the remaining problems is also insufficient.

Working with AI

We are studying in an age in which AI tools are widely available. The free versions of these tools are sufficient for the kinds of support expected in this course: asking questions, clarifying notation, generating simpler examples, checking reasoning, identifying missing steps, and improving the organisation of mathematical writing. Access to expensive software is not required.

AI should be used as an individual learning assistant. It may help transform rough calculations into clear mathematical prose, but it does not replace the student's mathematical work. Receiving a well-written answer is not the same as understanding it. The student remains responsible for checking every definition, calculation, assumption, conclusion, and formula included in the submitted note.

Use AI with a focused context. Work on one problem at a time and prepare one complete note for that problem. Do not paste an entire problem set into a single conversation and request all answers at once. Handling many unrelated tasks together encourages brief, generic, and mathematically weak notes. Focused work makes it easier to question each step, identify missing justifications, and produce a solution that can be understood and presented.

The one-problem-at-a-time principle does not mean that only one problem from the set should be prepared. Students are expected to repeat this focused process for every problem in the complete assigned set.

A polished AI-generated note is insufficient if the student cannot read its formulas, explain its terminology, justify its transitions, or present the argument independently. When using AI, ask not only for a solution, but also questions that deepen understanding:

- What does this formula mean?
- What mathematical object does each symbol represent?
- Why is this step valid?
- Which definition or theorem is being used?
- Are the assumptions of the theorem satisfied?
- Is there a simpler way to explain this transition?
- How can the result be checked?

- What should I be able to explain during class?

Preparing notes in this way is part of learning mathematics. It develops the ability to read notation, recognise mathematical objects, organise information, construct arguments, and communicate ideas precisely. Repeating this process for the complete problem set builds the habits of systematic and conscientious work that should accompany the student throughout university study and later professional life.

Student GitHub Workspace

The current instructions, problem templates, and technical workflow for student work are maintained in the public repository `dchorazkiewicz/Math_Problems_Repo`. Students should create a public fork of this repository and commit their notes to their own fork. The course repository is the common template; it is not the place where individual student solutions should be submitted.

The repository contains a short map of the working process:

- `README.md` explains the purpose and structure of the workspace and provides the weekly index;
- `SOURCE_MATERIAL.md` links the current lecture PDF and maps each week to the corresponding LaTeX problem source used by an AI assistant;
- `NOTE_GUIDELINES.md` defines the standard of a complete mathematical note;
- `AI_WORKFLOW.md` describes the one-problem-at-a-time workflow with AI, review, commits, presentation, feedback, and revision;
- `AGENTS.md` gives a connected AI assistant the operational rules for working safely in the student's fork;
- the `docs/` directory contains twelve weekly folders and one Markdown file for each problem.

The workspace uses Markdown for student notes, Git and GitHub for version history and public forks, MathJax for mathematical typesetting, MkDocs for the web presentation, and GitHub Actions for automatic validation and deployment. These instructions and templates may be updated independently of this PDF, so students should consult the repository for the current working procedure. The repository links back to this PDF and to its LaTeX sources. Students use the PDF as the readable course material containing the theory, definitions, examples, and problem sets. A connected AI assistant uses the current mapped LaTeX file as the exact machine-readable source of a selected problem statement.

Chapter 1

The Cartesian Plane

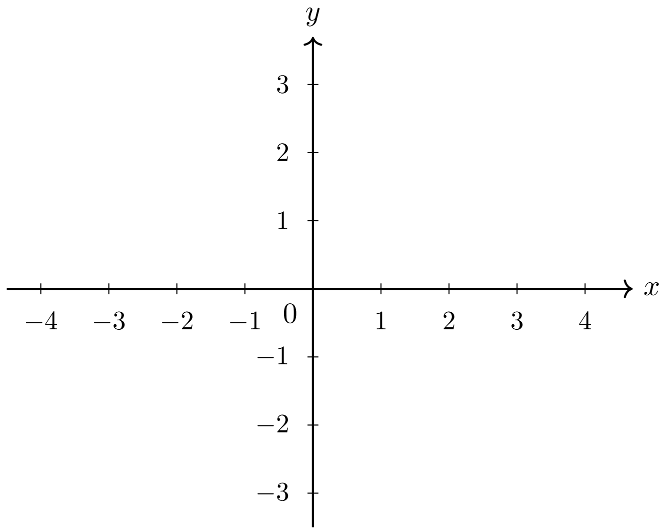
1.1 Points and Coordinates in the Plane

Before we introduce algebraic techniques, we need a place in which mathematical objects can live and be seen. The real number line is useful, but it allows us to move in only one direction. Many questions become clearer as soon as we can move independently in two directions and draw what is happening. For this reason, our first working environment will be the Cartesian plane.

The plane is simple enough to draw on paper, but already rich enough to contain points, curves, regions, distances, directions, equations, and transformations. Three-dimensional space will later extend many of the same ideas, but two dimensions allow us to see the essential structure without hiding it behind a complicated picture.

1.1.1 Two Number Lines Crossing

Take two copies of the real number line. Place one horizontally and the other vertically so that they meet at their zero points.



The horizontal line is called the *x-axis*, and the vertical line is called the *y-axis*. Their intersection is called the *origin*. We denote the origin by

$$O = (0, 0).$$

The axes give us two independent measurements. The first measurement tells us how far we move horizontally from the origin. The second tells us how far we move vertically.

Definition

A point in the Cartesian plane is described by an ordered pair of real numbers

$$(x, y).$$

The first number is the *x-coordinate*; the second is the *y-coordinate*.

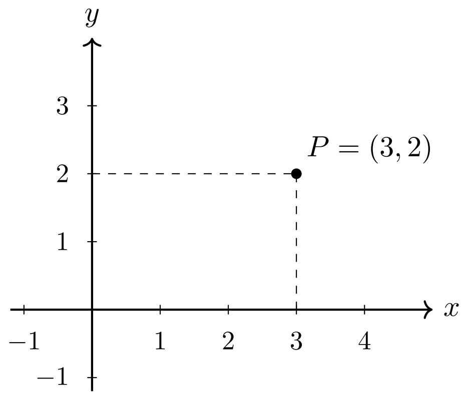
The word *ordered* is essential. The point $(2, 5)$ is generally not the same point as $(5, 2)$. The first number always refers to horizontal position, and the second always refers to vertical position.

1.1.2 From a Pair of Numbers to a Point

Consider the ordered pair

$$P = (3, 2).$$

To locate this point, begin at the origin. Move three units to the right, then two units upward. The endpoint is the point P.



The dashed lines are not part of the point. They only show how the coordinates are read from the axes. The vertical dashed line reaches the x-axis at 3, and the horizontal dashed line reaches the y-axis at 2.

Now consider

$$Q = (-2, 3).$$

The negative first coordinate tells us to move two units to the left. The positive second coordinate tells us to move three units upward.

A coordinate therefore contains both magnitude and direction:

- a positive x-coordinate means right of the y-axis;
- a negative x-coordinate means left of the y-axis;
- a positive y-coordinate means above the x-axis;
- a negative y-coordinate means below the x-axis.

1.1.3 From a Point to a Pair of Numbers

The same correspondence can be read in the opposite direction. Given a point in the plane, we determine its coordinates by projecting it onto the axes.

Example

Suppose a point lies four units to the left of the y-axis and one unit below the x-axis. Its coordinates are

$$(-4, -1).$$

The first coordinate records the horizontal position, and the second records the vertical position.

This gives a precise correspondence:

$$\text{point in the plane} \quad \longleftrightarrow \quad \text{ordered pair of real numbers.}$$

The geometry and the arithmetic describe the same object in two different languages. A point can be seen in a diagram, but it can also be stored and manipulated as a pair of numbers.

1.1.4 The Cartesian Plane as a Set

Because each point is represented by two real numbers, the entire Cartesian plane can be written as

$$\mathbb{R}^2 = \{(x, y) : x \in \mathbb{R},\ y \in \mathbb{R}\}.$$

The braces describe a set. The expression inside them says which objects are admitted to that set. We read the formula as follows:

> $\mathbb{R}^2$ is the set of all ordered pairs (x, y) such that x is a real number and y is a real number.

The superscript 2 does not mean that the real numbers are being squared. It tells us that two real coordinates are required to specify one point.

For example,

$$(1, 4), \qquad (-3, 0), \qquad \left(\frac{1}{2}, -7\right), \qquad (\sqrt{2}, \pi)$$

are all elements of $\mathbb{R}^2$.

1.1.5 Conditions and Sets of Points

A formula involving x and y may be used in two different ways. First, it may appear only as a *condition*. For example,

$$y = 2$$

is a relation between two numbers. By itself, it tells us which pairs of numbers satisfy the relation, but it does not yet introduce a name for the collection of all such pairs.

To define a geometric object as a subset of the plane, we write the entire set:

$$L = \{(x, y) \in \mathbb{R}^2 : y = 2\}.$$

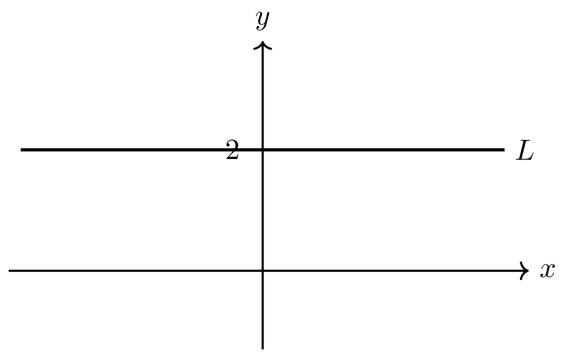
This is read as follows:

> L is the set of all points (x, y) in the Cartesian plane such that $y = 2$.

The colon means "such that." A vertical bar may be used instead:

$$\{(x, y) \in \mathbb{R}^2 : y = 2\} = \{(x, y) \in \mathbb{R}^2 \mid y = 2\}.$$

The condition $y = 2$ selects precisely those points whose second coordinate is 2. Geometrically, they form a horizontal line.



Several particular elements of this set are

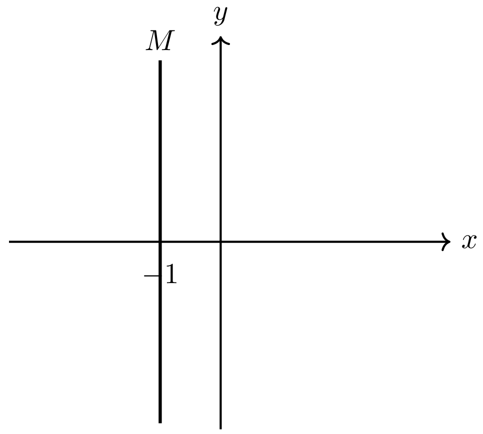
$$(0,2), \quad (1,2), \quad (-3,2), \quad \left(\frac{1}{2},2\right).$$

They are examples of points in L, but no finite list of examples is the same thing as the complete definition of L.

Similarly, the set

$$M = \{(x,y) \in \mathbb{R}^2 : x = -1\}$$

is the vertical line through $x = -1$.



This distinction will be used throughout the course:

$$\boxed{\text{condition on coordinates}} \qquad \text{versus} \qquad \boxed{\text{set of all points satisfying the condition}}.$$

1.1.6 The Axes and Quadrants as Subsets of the Plane

The coordinate axes themselves are subsets of $\mathbb{R}^2$. The x-axis is

$$X = \{(x,y) \in \mathbb{R}^2 : y = 0\},$$

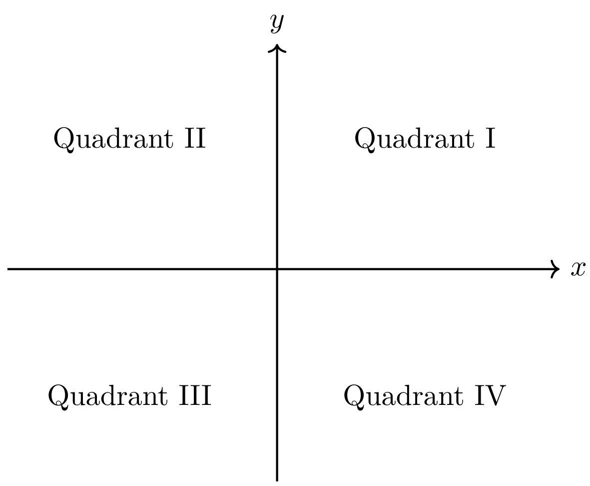
and the y-axis is

$$Y = \{(x,y) \in \mathbb{R}^2 : x = 0\}.$$

Their only common point is the origin:

$$X \cap Y = \{(0,0)\}.$$

The two axes divide the plane into four regions called *quadrants*.



For example, the first quadrant is the set

$$Q_1 = \{(x, y) \in \mathbb{R}^2 : x > 0 \text{ and } y > 0\}.$$

The remaining quadrants are described similarly:

$$Q_2 = \{(x, y) \in \mathbb{R}^2 : x < 0 \text{ and } y > 0\},$$
$$Q_3 = \{(x, y) \in \mathbb{R}^2 : x < 0 \text{ and } y < 0\},$$
$$Q_4 = \{(x, y) \in \mathbb{R}^2 : x > 0 \text{ and } y < 0\}.$$

Points on the axes do not belong to any quadrant because at least one of their coordinates is zero.

1.1.7 A Curved Set of Points

Consider now the condition

$$x^2 + y^2 = 4.$$

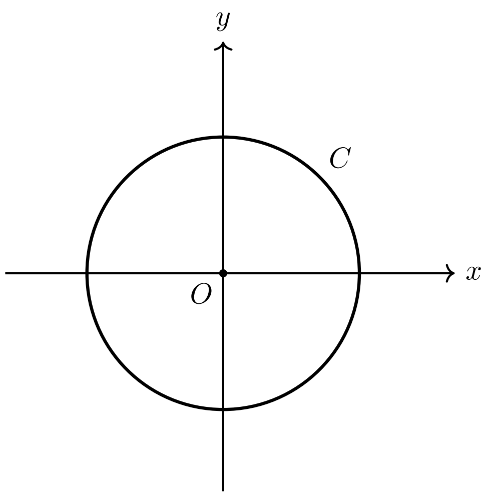
Some points satisfy it:

$$(2, 0), \quad (-2, 0), \quad (0, 2), \quad (0, -2).$$

Other points do not. The complete geometric object is the subset

$$C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}.$$

We read this as "the set of all points (x, y) in the plane such that $x^2 + y^2 = 4$." Geometrically, C is the circle of radius 2 centred at the origin.



The equation is the condition. The set-builder expression states explicitly which ambient space we are working in, what the elements are, and which condition they must satisfy. The diagram then shows the geometric meaning of that set.

This gives one of our main working principles:

$$\text{condition on coordinates} \quad \longrightarrow \quad \text{subset of } \mathbb{R}^2 \quad \longrightarrow \quad \text{geometric interpretation.}$$

The Cartesian plane therefore gives us more than a picture. It provides a common language in which numerical information, algebraic conditions, sets, and geometric objects can be compared directly.

1.2 Another Way to Describe a Point: Polar Coordinates

The Cartesian description of a point uses two signed distances:

$$P = (x, y).$$

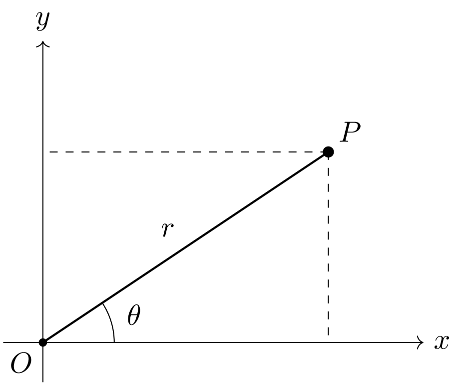
This is natural when horizontal and vertical directions are important. It is not, however, the only possible way to describe position in the plane. We may instead ask two different questions:

1. How far is the point from the origin?
2. In which direction do we move from the origin to reach it?

These questions lead to *polar coordinates*. A point is described by a pair

$$(r, \theta),$$

where $r \geq 0$ is the distance from the origin and θ is the angle measured from the positive x-axis.



The two descriptions refer to the same point. The right triangle in the figure gives

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Conversely, if the Cartesian coordinates are known, then

$$r = \sqrt{x^2 + y^2}.$$

The angle must be chosen so that its sine and cosine have the correct signs. For this reason, the formula $\theta = \arctan(y/x)$ must be used with care: it does not by itself distinguish all four quadrants.

Example

Consider the point whose polar coordinates are

$$(r, \theta) = \left(2, \frac{\pi}{3}\right).$$

Its Cartesian coordinates are

$$x = 2\cos\frac{\pi}{3} = 1, \qquad y = 2\sin\frac{\pi}{3} = \sqrt{3}.$$

Thus the same point may be written as

$$P = \left(1, \sqrt{3}\right)$$

in Cartesian coordinates and as

$$P = \left(2, \frac{\pi}{3}\right)$$

in polar coordinates.

Polar coordinates are especially convenient when distance from the origin is the main geometric feature. Let

$$C_R = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\}.$$

This is the set of all points in the Cartesian plane whose coordinates satisfy the condition

$$x^2 + y^2 = R^2.$$

Geometrically, C_R is the circle of radius R centred at the origin. After substituting

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

the Cartesian condition becomes

$$r^2\cos^2\theta + r^2\sin^2\theta = R^2.$$

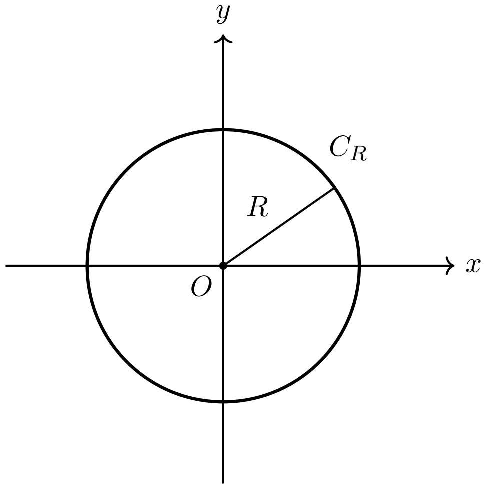
Since $\cos^2\theta + \sin^2\theta = 1$, this reduces to

$$\boxed{r = R}.$$

The set C_R has not changed. The conditions

$$x^2 + y^2 = R^2 \qquad \text{and} \qquad r = R$$

describe the same geometric subset of the plane, but they use different coordinates. In Cartesian coordinates we express a relation between horizontal and vertical position. In polar coordinates we state directly that every point of the set has the same distance from the origin.



Remark

Polar coordinates are not unique. The angles θ and $\theta + 2\pi$ describe the same direction, so the pairs

$$(r, \theta) \qquad \text{and} \qquad (r, \theta + 2\pi)$$

represent the same point in the plane. At the origin, where $r = 0$, the angle is irrelevant. This is a first example of an important distinction: a geometric object may be unique even when its coordinate description is not.

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